

Supplementary Material for

Beyond Pre-Post Comparisons: A Comprehensive Effect Size Framework for Intensive Longitudinal Interventions via Time-Varying Modeling

This supplement defines the decay effect sizes that constitute the fourth dimension (study phase) of the proposed framework. The metrics apply to ILI designs that include a post-intervention phase, that is, a period of continued intensive longitudinal assessment after the intervention has ended. Let the post-intervention phase have length T_3 , and let $\mu_3(t)$, $\varphi_3^*(t)$, and $V_3(t)$ denote the time-varying mean, standardized autoregressive (AR) coefficient, and model-implied variance during this phase, estimated with the same multilevel TV-AR models introduced in the main text.

The decay effect sizes follow exactly the same logic as the intervention effect sizes. Only the counterfactual changes. Whereas the intervention effects compare the intervention phase against a pre-intervention reference, the decay effects compare the post-intervention phase against a sustained-effect reference, namely the parameter value reached at the end of the intervention phase. Formally, the counterfactual trajectories are $\mu_3'(t) = \mu_2(T_2)$, $\varphi_3^{*'}(t) = \varphi_2^*(T_2)$, and $V_3'(t) = V_2(T_2)$ for all t in the post-intervention phase. These references represent what would be expected if the state achieved by the end of the intervention were fully maintained. Deviations from them therefore quantify the loss (or continued growth) of the intervention effect after support is withdrawn. For an outcome such as negative affect, where the intervention is expected to lower the mean, a positive mean decay effect indicates a rebound toward pre-intervention levels, a value near zero indicates maintenance, and a negative value indicates continued improvement. Standardization uses the same natural-state standard

deviation σ (or σ_i) as in the main text, so that intervention and decay effects are expressed on the same scale, and the variance-based metrics again use the log variance ratio.

Table S1 lists the decay metrics for the first model specification (time-varying latent mean with a time-invariant AR coefficient), paralleling Table 1 of the main text. Table S2 lists the decay metrics for the second specification (time-varying intercept and AR coefficient), paralleling Table 2. All metrics are computed per posterior draw, which yields point estimates (posterior medians) and 95% credible intervals in the same way as for the intervention effects. Between-group decay effects contrast the two groups' within-group decay effects and thereby test whether the treatment group's changes after the intervention differ from the changes observed in the control group over the same period.

Two remarks may guide applied use. First, the sustained-effect reference is not the only defensible choice. Comparing the post-intervention phase against the pre-intervention reference instead quantifies the residual benefit that remains after the intervention, and the two comparisons are complementary, because a small residual benefit can arise either from a weak intervention effect or from strong decay. Reporting the intervention effect, the decay effect, or both, should be guided by the substantive question. Second, the decay effect sizes require fitting the multilevel TV-AR model to the post-intervention phase itself, so the sample size requirements of these models apply to this phase as well. Previous simulation work recommends at least 100 participants and at least 100 time points for accurate estimation of the time-varying parameters (Luo et al., 2026). Researchers who intend to quantify decay should therefore recruit a sufficient number of participants in each group and plan a sufficiently long post-intervention assessment segment.

Table S1. Decay effect size metrics based on a multilevel TV-AR model with a time-varying intercept and a time-varying AR coefficient.

Subject	Temporal	Effect size metric	Definition equation
Population-average	Integrated	Population-average unstandardized integrated decay effect	$Unstd. Decay^{(W)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\mu_3(t) - \mu'_3(t)] dt$
		Population-average standardized integrated decay effect	$Std. Decay^{(W)} = \{\frac{1}{T_3} \int_{t=1}^{T_3} [\mu_3(t) - \mu'_3(t)] dt\} / \sigma$
	Time-specific	Population-average standardized time-varying decay effect	$Std. Decay^{(W)}(t) = [\mu_3(t) - \mu'_3(t)] / \sigma$
Individual-specific	Integrated	Individual-specific unstandardized integrated decay effect	$Unstd. Decay_i^{(W)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\mu_{3i}(t) - \mu'_{3i}(t)] dt$
		Individual-specific standardized integrated decay effect	$Std. Decay_i^{(W)} = \{\frac{1}{T_3} \int_{t=1}^{T_3} [\mu_{3i}(t) - \mu'_{3i}(t)] dt\} / \sigma_i$
	Time-specific	Individual-specific standardized time-varying decay effect	$Std. Decay_i^{(W)}(t) = [\mu_{3i}(t) - \mu'_{3i}(t)] / \sigma_i$
Between-group	Integrated	Between-group standardized integrated decay effect	$Std. Decay^{(B)} = Std. Decay_{g=T}^{(W)} - Std. Decay_{g=C}^{(W)}$
	Time-specific	Between-group standardized time-varying decay effect	$Std. Decay^{(B)}(t) = Std. Decay_{g=T}^{(W)}(t) - Std. Decay_{g=C}^{(W)}(t)$

Table S2. Decay effect size metrics based on a multilevel TV-AR model with a time-varying intercept and a time-varying AR coefficient.

Subject	Parameter	Temporal	Effect size metric	Definition equation
Population-average	Mean (Attractor)	Integrated	Population-average unstandardized integrated decay effect	$Unstd.Decay^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\mu_3(t) - \mu'_3(t)] dt$
			Population-average standardized integrated decay effect	$Std.Decay^{(w)} = \{\frac{1}{T_3} \int_{t=1}^{T_3} [\mu_3(t) - \mu'_3(t)] dt\} / \sigma$
		Time-specific	Population-average standardized time-varying decay effect	$Std.Decay^{(w)}(t) = [\mu_3(t) - \mu'_3(t)] / \sigma$
	Autoregressive coefficient	Integrated	Population-average standardized integrated autoregressive decay effect	$Std.Decay_{\varphi}^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\varphi_3^*(t) - \varphi_3^{*'}(t)] dt$
		Time-specific	Population-average standardized time-varying autoregressive decay effect	$Std.Decay_{\varphi}^{(w)}(t) = \varphi_3^*(t) - \varphi_3^{*'}(t)$
	Variance	Integrated	Population-average unstandardized integrated variance decay effect	$Unstd.Decay_V^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [V_3(t) - V'_3(t)] dt$
			Population-average standardized integrated variance decay effect	$Std.Decay_V^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} \log [V_3(t) / V'_3(t)] dt$
		Time-specific	Population-average standardized time-varying variance decay effect	$Std.Decay_V^{(w)}(t) = \log [V_3(t) / V'_3(t)]$

Individual-specific	Mean (Attractor)	Integrated	Individual-specific unstandardized integrated decay effect
			Individual-specific standardized integrated decay effect
		Time-specific	Individual-specific standardized time-varying decay effect
	Autoregressive coefficient	Integrated	Individual-specific standardized integrated autoregressive decay effect
		Time-specific	Individual-specific standardized time-varying autoregressive decay effect
	Variance	Integrated	Individual-specific unstandardized integrated variance decay effect
			Individual-specific standardized integrated variance decay effect
		Time-specific	Individual-specific standardized time-varying variance decay effect

$$Unstd.Decay_i^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\mu_{3i}(t) - \mu'_{3i}(t)] dt$$

$$Std.Decay_i^{(w)} = \{\frac{1}{T_3} \int_{t=1}^{T_3} [\mu_{3i}(t) - \mu'_{3i}(t)] dt\} / \sigma_i$$

$$Std.Decay_i^{(w)}(t) = [\mu_{3i}(t) - \mu'_{3i}(t)] / \sigma_i$$

$$Std.Decay_{\phi i}^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [\varphi_{3i}^*(t) - \varphi'_{3i}(t)] dt$$

$$Std.Decay_{\phi i}^{(w)}(t) = \varphi_{3i}^*(t) - \varphi'_{3i}(t)$$

$$Unstd.Decay_{Vi}^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} [V_{3i}(t) - V'_{3i}(t)] dt$$

$$Std.Decay_{Vi}^{(w)} = \frac{1}{T_3} \int_{t=1}^{T_3} \log [V_{3i}(t) / V'_{3i}(t)] dt$$

$$Std.Decay_{Vi}^{(w)}(t) = \log [V_{3i}(t) / V'_{3i}(t)]$$

Between-group	Mean (Attractor)	Integrated	Between-group standardized integrated decay effect	$Std. Decay^{(B)} = Std. Decay_{g=T}^{(W)} - Std. Decay_{g=C}^{(W)}$
		Time-specific	Between-group standardized time-varying decay effect	$Std. Decay^{(B)}(t) = Std. Decay_{g=T}^{(W)}(t) - Std. Decay_{g=C}^{(W)}(t)$
	Autoregressive coefficient	Integrated	Between-group standardized integrated autoregressive decay effect	$Std. Decay_{\phi}^{(B)} = Std. Decay_{\phi,g=T}^{(W)} - Std. Decay_{\phi,g=C}^{(W)}$
		Time-specific	Between-group standardized time-varying autoregressive decay effect	$Std. Decay_{\phi}^{(B)}(t) = Std. Decay_{\phi,g=T}^{(W)}(t) - Std. Decay_{\phi,g=C}^{(W)}(t)$
	Variance	Integrated	Between-group standardized integrated variance decay effect	$Std. Decay_V^{(B)} = Std. Decay_{V,g=T}^{(W)} - Std. Decay_{V,g=C}^{(W)}$
		Time-specific	Between-group standardized time-varying variance decay effect	$Std. Decay_V^{(B)}(t) = Std. Decay_{V,g=T}^{(W)}(t) - Std. Decay_{V,g=C}^{(W)}(t)$